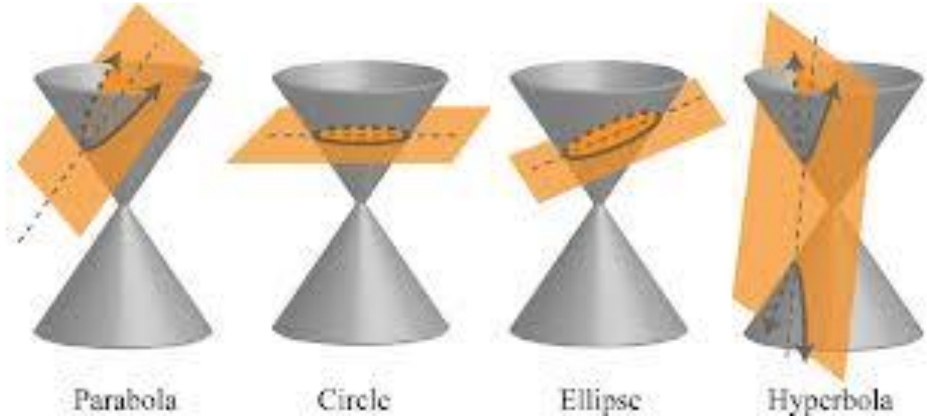
MTH 201 Chapter 5: Parametric/Polar Curves & Conics

5.4.3 Conic Sections (Hyperbolas)

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.

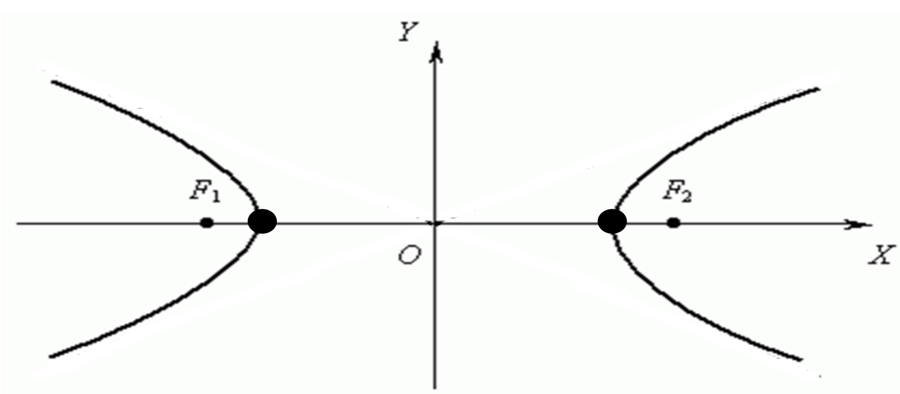


Definition: A **hyperbola** is the set of all points in a plane such that the difference between the distances from two fixed points (foci) is constant.

*Note the subtle difference between this definition and the definition for an ellipse which has the *sums* of differences from two fixed points as constant.*

The Equation of a Hyperbola: (opening left or right)

The equation of a hyperbola with center at the origin

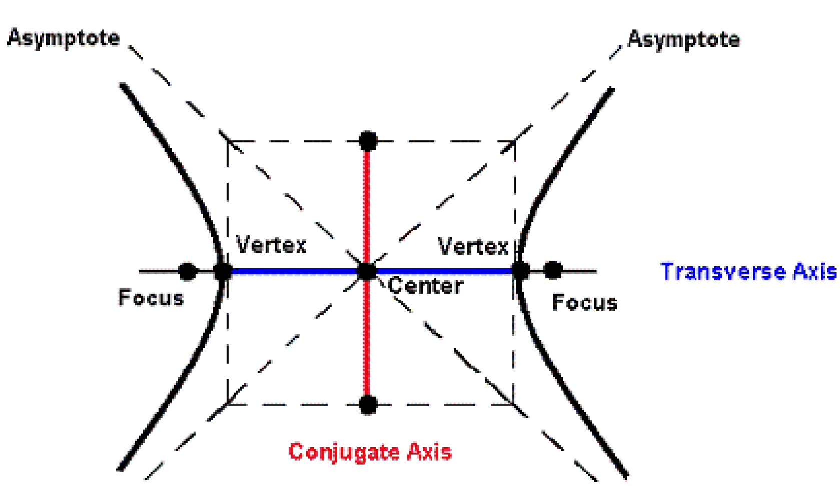


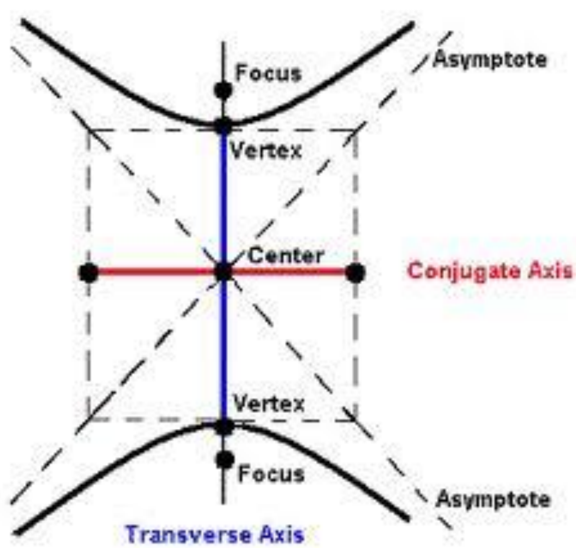
The Equation of a Hyperbola: (opening up or down)

The equation of a hyperbola with center at the origin

Note: With hyperbolas it is not important whether a or b is the larger value. We always think of a as the value beneath y and b as the value beneath x.

Graphing a Hyperbola





Example 1: Determine the foci, equations of the asymptotes, and sketch the graph of

$$\frac{x^2}{36} - \frac{y^2}{9} = 1$$

Example 2: Determine the foci, equations of the asymptotes, and sketch the graph of

$$4y^2 - 9x^2 = 36$$

Example 3: Determine the foci, equations of the asymptotes, and sketch the graph of

$$\frac{(x-2)^2}{9}-\frac{(y+1)^2}{4}=1$$

Example 4: Find the equation of a hyperbola whose transverse axis had endpoints $(0,\pm 4)$ and whose conjugate axis has endpoints $(\pm 2,0)$.

Example 5: Find the equation of the hyperbola with asymptotes $y=\pm 4x$ and vertices $(\pm 3,0)$.

Recapping Section 5.4

Example 6:

(a) Find an equation for the parabola with vertex $(2,3)$, axis of symmetry $x=2$, and which contains the point $(1,5)$.

(b) Find the vertex/vertices of the conic determined by the equation $9x^2-4y^2-72x+8y+176=0$.

(c) Find an equation of the ellipse with foci $(2,-2)$, $(4,-2)$ and vertices $(1,-2)$, $(5,-2)$.